JAYASURIYA S

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EDUCATION

B.E. Computer Science & Engineering, Sri Eshwar College of Engineering (2024–2025) | CGPA: 8.20
St. Mary's Anglo Indian Higher Secondary School | HSC – 96% | SSLC – 98%

PROJECTS

URBAN-PRISM – AI-Assisted Grievance Intelligence Portal (Repo)

• Developed an AI-powered grievance intelligence system that converts multilingual citizen complaints into structured urban insights using automated intake and processing.
• Implemented geospatial clustering and asset mapping to identify location-based issue patterns and high-risk zones.
• Designed a risk-based prioritization framework to classify and rank complaints, enabling data-driven urban decision-making.

LUNEX – Smart Laundry Management System (RFID + ESP32) (Repo)

• Built a full-stack smart laundry management platform integrating web-based slot booking with RFID authentication and ESP32 hardware control.
• Implemented secure access validation and real-time booking workflows using RESTful APIs and backend logic.
• Demonstrated IoT–software integration to optimize shared resource utilization and reduce scheduling conflicts in hostel environments.

MarketMate – MERN-Based Local Price Comparison Platform (Repo)

• Developed a full-stack MERN application enabling local shopkeepers to list products and users to compare prices across nearby stores.
• Implemented role-based authentication with admin approval workflows and product management dashboards.
• Integrated search, filtering, sorting, and dynamic UI state management to enhance user experience and usability.

AI-Powered Wildlife Poaching Detection System (Repo)

• Developed an end-to-end machine learning system integrating YOLOv8 object detection on satellite imagery with Isolation Forest–based anomaly detection on GPS collar data.
• Engineered movement-based features (speed, acceleration, direction change) to detect abnormal animal behavior, achieving a 78% F1-score.
• Built a multi-modal alert fusion engine with interactive Folium-based geospatial visualization to generate prioritized alerts for scalable monitoring and rapid response.

SafeTrack – Urban Safety Analytics System (Repo)

- Designed and implemented a **machine-learning geospatial analytics model** to classify urban safety risk levels (Low/Medium/High) using Indian crime data.
- Applied **LightGBM classification** with feature engineering on time-of-day, crime category, and city-level risk indicators to achieve **75–85% accuracy**.
- Developed the system for **ethical urban planning and safety research**, aligned with **UN SDG-5 and SDG-11**.

SKILLS

Programming Languages: C, C++, Python, Java
Web Development & Technologies: HTML, CSS, JavaScript, MERN Stack (Basics): React.js, Node.js, Express.js, Spring Boot
Databases: MySQL, MongoDB
Cloud & Deployment: AWS (EC2, S3), Vercel, Render
Tools & Platforms: Git, GitHub, VS Code, WordPress, Canva, MS Word, Excel, PowerPoint

CERTIFICATIONS

Certifications: DSA using C/C++ (Udemy), Java Basics (Oracle), Java on Azure (Microsoft), Full-Stack AI Projects (Udemy)

CODING PROFILES

SkillRack: 840+ problems solved | **239+ Bronze Awards** | Link | **LeetCode:** 50+ problems solved | Link

COMPETITIONS AND TECHNICAL ACTIVITIES

- Secured the **Best Implementation Award** at **Vel Ideaforge 2K26**, Vel Tech College, for developing *Urban Prism*, receiving a ₹5000 cash prize.
- **Hack ML Hackathon:** Secured **1st Place** for developing *SafeTrack*, an AI-based urban safety analytics system addressing crimes against women using machine learning.
- Participated in **college-level hackathons and technical project reviews**, presenting AI/ML and full-stack solutions.

ENTREPRENEURIAL EXPERIENCE

Founder – Havox. Founded a streetwear clothing brand and built a production-ready WordPress e-commerce website with Razorpay payment integration. (havox.in)

LEADERSHIP ROLES

Secretary, Mentoring Vertical – Sri Eshwar College of Engineering
School Pupil Leader – St. Mary's Anglo Indian Higher Secondary School

RELEVANT COURSEWORK

Data Structures and Algorithms, Object Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering